

PILE-INSTALLATION PERFORMANCE FOR FOUR TLP'S IN THE GULF OF MEXICO

Shell has installed four tension-leg platforms (TLP's) in the U.S. Gulf of Mexico. Installation of the first TLP began in December 1993 at the Auger prospect in a water depth of 2,860 ft. The foundation consists of four individual templates, each secured to the seafloor by four 72-in.-diameter vertical piles driven to a penetration of 374 ft below the seafloor. Three tendons were attached to each template.

The second TLP was installed at the Mars prospect during April and May 1996 in a water depth of 2,940 ft. Three 84-in.-diameter piles per corner were driven to a penetration of 375 ft. This TLP was the first to have tendons attached to the piles directly, eliminating the need for seafloor templates. Subsequent TLP's (Ram-Powell and Ursa) used the direct-connection method to connect tendons to piles.

The third TLP was installed at the Ram-Powell prospect in May 1997. Three 84-in.-diameter piles per corner were driven to a penetration of 328 ft in a water depth of 3,214 ft. The fourth TLP was installed at the Ursa prospect in November 1998 in a water depth of 3,800 ft. Four 96-in.-diameter piles per corner were driven to a penetration of 396 ft.

SOIL PROPERTIES

Normal or slightly overconsolidated clay underlies each TLP site. Drilling/sampling and in-situ testing were conducted from dynamically positioned vessels. For installation considerations, the remolded-soil properties are important parameters. The skin resistance for self-weight pile penetration and the resistance to driving are controlled by the remolded-soil properties.

For Auger, the miniature-vane (MV) remolded shear strengths were slightly lower than the unconsolidated/undrained

triaxial (UUT) remolded shear strengths. The Mars boring was drilled to a penetration of 593 ft. At Mars, the UUT tests were not begun until a penetration of 115.8 ft. The Ram-Powell boring was drilled to a penetration of 488 ft. The UUT remolded strengths are much higher than the MV strengths at this site. The Ursa boring was drilled to a penetration of 451 ft below the seafloor. The UUT remolded strengths are slightly higher than the MV strengths at this site.

SELF-WEIGHT PENETRATION

Column buckling caused by the dead weight of an underwater hammer placed on top of a tilted, self-weight-penetrated pile is a design case for TLP's that required increased wall thickness in some sections of the pile. The Auger TLP, with its seafloor templates, provided the largest bending moments at the bottom of the free-standing pile section after self-weight penetration (SWP) because of the constraints provided by the pile sleeves. Attaching the tendons directly to the piles at Mars, Ram-Powell, and Ursa were free-head cases where column buckling was checked, but the controlling case usually was limiting the lateral pile-driving deflections. Thus, SWP was an important parameter in the pile design.

The general method to calculate the SWP assumed that the resistance to penetration was provided by skin friction along the outside and inside skin friction provided by the shoe and the steel-end-bearing area at the base of the pile. The remolded-soil strength was used for skin friction, and the undisturbed strength was used for end bearing. The energy to penetrate was provided by the underwater weight of the pile and the internal lifting tool (ILT) used to lower the pile to the seafloor.

In general, the lowering procedure from the mudline to SWP depth was similar for all piles. All the piles self-penetrated at approximately 1 ft/sec, one-half the crane's maximum velocity. The range in SWP's at every site, except Auger, was relatively small (Mars: 117 to 120 ft; Ram-Powell: 111 to 113 ft; and Ursa: 123 to 128 ft). The range at Auger of 87 to 98 ft was, in part,

caused by variations in crane-lowering speeds. At Ursa, pile-drive depths within a corner varied by less than 3 ft, indicating that the lateral variation in remolded-soil strengths at each site is relatively small.

Pile SWP can be calculated conservatively by use of UUT remolded strengths applied as skin friction both to the outside of the pile and to the inside area of the shoe. The undisturbed shear strength should be applied to the steel-end-bearing area of the shoe. The driving force should include the underwater weight of both the pile and the ILT.

PILE DRIVING

For most driveability analyses, the aim is to size the hammer to ensure that the piles do not meet refusal and that the dynamic driving stresses are within allowable limits. An additional requirement for the TLP piles is to minimize the driving-induced-fatigue damage so that the pile retains reserve fatigue life during the life of the structure. In general, driving-induced damage is more sensitive to the magnitude of the stress waves traveling within the pile than to the total number of blows by the hammer. Thus, a conservative prediction that results in an overestimate of blows could result in a pile with more fatigue damage than predicted. This overestimate is because the actual stress wave traveling within the pile has a stronger tension component returning up the pile than predicted in the preinstallation driveability studies. To minimize driving-induced energy, the upper sections are driven at a reduced energy level such that the tension component of the stress wave is minimized. All piles were driven with an underwater hydraulic hammer.

Average blow counts for the last foot driven ranged from 32 to 38, except Ursa with an average of 76. The average total blows to drive each pile ranged from 5,136 to 5,880, except Ursa with an average of 9,528. The maximum blow count can be estimated conservatively by use of the MV remolded strength as the skin friction. For fatigue calculations, closer predictions were made with the interpreted remolded strengths along with Roussel's damping and quake

This article is a synopsis of paper OTC 10826, "Pile-Installation Performance for Four TLP's in the Gulf of Mexico," by Earl H. Doyle, Consultant, originally presented at the 1999 Offshore Technology Conference, Houston, 3-6 May.

parameters that also are based on remolded strengths. More precise blow-count predictions for fatigue analyses may require that piles be instrumented to obtain the necessary parameters for future analyses.

The Mars and Ram-Powell installations experienced significant hammer downtime. The longest interruption occurred at Mars when driving was interrupted for 4.7 days to repair the hammer. An analysis using 0.35 times the undrained shear strength approximates the driving record before the breakdown. Following the breakdown, the skin friction had to be increased to 0.45 times the undrained shear strength to follow the post-breakdown driving record. Use of a doubly redundant hammer system at Ursa reduced downtime considerably. In addition, reliability of the hammer system improved to the point that the final eight piles were driven without any breakdowns or maintenance trips to the surface.

PILE LOCATIONS AND TILT

The Auger TLP used individual templates at each corner. The tendon-fatigue requirements were such that the template tilt needed to be minimized. To ensure level templates, a closely gridded bathymetry survey

was performed on the seafloor beneath each template location. The template mud mats were adjusted to conform to the seafloor topography. Initial settlement analyses were conducted to predict the settlement on touchdown. The observed 5- to 10-in. elastic settlement was within the range of the analytical techniques used. The actual seafloor tilt varied between 0.9 and 1.4°, which was within design tolerance.

The Mars, Ram-Powell, and Ursa installations involved lowering a single pile to the seafloor and accurately installing each pile within a closely confined target circle with a minimum pile tilt. Off-target piles cause fatigue in the bottom tendon connector. In addition, pile tilt affects the column-buckling characteristics when the hammer is placed on the pile following SWP. For Mars, a 2° tilt would have increased the maximum design bending moments by more than 30%. Thus, it is important to minimize the off-target range and tilt and to install the piles with as small a variation in these quantities as possible so that reasonable tilt and range criteria can be developed. The final average tilt ranged from 0.21 to 0.41°. The average final off-target distances varied from 1.00 to 1.08 ft. These

remarkably small numbers were achieved by careful planning and attention to detail by the installation and survey teams. In addition, these achievements were completed in a relatively short time period. For Ursa, the average time between lifting the pile from the pile barge and completion of SWP was 7½ hours for the last twelve piles. (No mechanical or weather time is included in this number.)

The Auger template installation showed that tilt can be minimized if careful attention is given to the seafloor topography and the mud mats are installed on each template to accommodate a differential seafloor slope. The Mars, Ram-Powell, and Ursa installations showed that piles can be positioned on the seafloor precisely with a minimum of tilt in deep water. **JPT**

Please read the full-length paper for additional detail, illustrations, and references. The paper from which the synopsis has been taken has not been peer reviewed. Copyright 1999 Offshore Technology Conference.